

Combined score plus top singular directions. This variant of the combined-score method splits the retained sketch budget r into two parts,

$$r = r_1 + r_2, \quad r_1, r_2 \geq 0. \quad (1)$$

The first r_1 retained directions are obtained by sequentially maximizing the combined score of the previous note. The remaining r_2 directions are the top right singular vectors of the stacked operator

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}, \quad (2)$$

restricted to the orthogonal complement of the score-selected directions. This gives a hybrid basis: the combined score targets entropy-corrected leverage, while the singular-value part guarantees coverage of the leading quadratic energy of M_w .

Setup and notation. Let $A \in \mathbb{R}^{m \times d}$ be the full data matrix, let $A_w \in \mathbb{R}^{k \times d}$ be the current window with $k = \text{row}(A_w)$, let $B \in \mathbb{R}^{r \times d}$ be the sketch carried from the previous window, and let $R \in \mathbb{R}^{s \times d}$ be the old row memory. Write

$$n_{\text{seen}} := \# \text{ raw rows represented by } B \text{ and } R \text{ before this window}, \quad (3)$$

$$n := \text{total } \# \text{ rows in the full matrix or stream}, \quad (4)$$

$$p_w := \frac{n_{\text{seen}} + k}{n}, \quad (5)$$

$$c_w := \frac{\log p_w}{2 \log(s + k)}, \quad (6)$$

$$E_w(v) := \|Bv\|_2^2 + \|A_w v\|_2^2, \quad (7)$$

$$C_w := \begin{pmatrix} R \\ A_w \end{pmatrix} \in \mathbb{R}^{(s+k) \times d}. \quad (8)$$

The combined score of the previous note is defined by

$$\text{score}_w(v) := E_w(v) p_w^{-h_2^{(R,w)}(v)/(2 \log(s+k))}, \quad (9)$$

with pooled entropy proxy

$$h_2^{(R,w)}(v) = -\log \left(\frac{\|C_w v\|_4^4}{\|C_w v\|_2^4} \right). \quad (10)$$

Assume $s + k > 1$ and $0 < p_w \leq 1$ when the correction is used. If no row memory or previous sketch is available, take $R = \emptyset$, $s = 0$, and $B = 0$.

Closed-form rewrite of the entropy multiplier. The entropy multiplier can be turned into a pure monomial in the ℓ_2 and ℓ_4 norms of $C_w v$. Write

$y := C_w v$. Substituting $h_2^{(R,w)}(v) = -\log(\|y\|_4^4/\|y\|_2^4)$ into the multiplier gives

$$p_w^{-h_2^{(R,w)}(v)/(2\log(s+k))} = \exp\left(-\frac{h_2^{(R,w)}(v)}{2\log(s+k)} \log p_w\right) \quad (11)$$

$$= \exp\left(-c_w h_2^{(R,w)}(v)\right) \quad (12)$$

$$= \exp\left(c_w \log \frac{\|y\|_4^4}{\|y\|_2^4}\right) \quad (13)$$

$$= \left(\frac{\|y\|_4^4}{\|y\|_2^4}\right)^{c_w}, \quad (14)$$

where the second line uses $c_w = \log p_w/(2\log(s+k))$ and the third line uses $-h_2^{(R,w)}(v) = \log(\|y\|_4^4/\|y\|_2^4)$.

Because $y = C_w v$ is the stacked vector $(Rv; A_w v)$,

$$\|y\|_2^2 = \|Rv\|_2^2 + \|A_w v\|_2^2, \quad \|y\|_4^4 = \|Rv\|_4^4 + \|A_w v\|_4^4, \quad (15)$$

so the entropy multiplier admits the explicit form

$$\phi_w(v) := p_w^{-h_2^{(R,w)}(v)/(2\log(s+k))} = \left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2}\right)^{c_w}. \quad (16)$$

The combined score is therefore

$$\text{score}_w(v) = E_w(v) \phi_w(v) = (\|Bv\|_2^2 + \|A_w v\|_2^2) \left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2}\right)^{c_w}. \quad (17)$$

This is the form used in Step 1 below. Note $c_w \leq 0$ since $p_w \leq 1$ and $s+k > 1$, so $\phi_w(v)$ increases as the pooled responses become more entropic (less concentrated).

Budget split. Fix integers $r_1, r_2 \geq 0$ with $r_1 + r_2 = r$. The allocation can be a fixed ratio (for example $r_1 = \lceil r/2 \rceil$) or data-driven. A natural data-driven rule is to set r_1 equal to the number of combined-score local maxima whose relative score gain exceeds a tolerance, and to set $r_2 = r - r_1$. The construction below does not depend on how the split is made; it requires only that $r_1 + r_2 = r$.

Step 1: combined-score directions. Compute r_1 orthonormal directions $v_1, \dots, v_{r_1}$ sequentially by solving, for $j = 1, \dots, r_1$,

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j} \text{score}_w(v). \quad (18)$$

Use the projected full-gradient ascent of the previous note: with $Q_j := [v_1, \dots, v_{j-1}]$ and tangent projection $(I - Q_j Q_j^\top - vv^\top)$, take backtracking gradient steps on score_w and re-project onto the feasible sphere. Collect the result as

$$V_1 := [v_1, \dots, v_{r_1}] \in \mathbb{R}^{d \times r_1}. \quad (19)$$

If $r_1 = 0$, set V_1 to the empty matrix.

Step 2: top right singular vectors of sketch plus window, orthogonal to V_1 . Let $P_1 := I - V_1 V_1^\top$ be the orthogonal projector onto the complement of $\text{span}(V_1)$. Consider the restricted operator

$$M_w P_1 = M_w - M_w V_1 V_1^\top. \quad (20)$$

Compute the top r_2 right singular vectors of $M_w P_1$:

$$V_2 := [u_{r_1+1}, \dots, u_{r_1+r_2}] \in \mathbb{R}^{d \times r_2}, \quad (21)$$

where $u_{r_1+1}, \dots, u_{r_1+r_2}$ are unit-norm, mutually orthogonal, and orthogonal to every column of V_1 , and where the ordering is by non-increasing singular value of $M_w P_1$. If $r_2 = 0$, set V_2 to the empty matrix.

Equivalently, V_2 is the solution of

$$\max_{V \in \mathbb{R}^{d \times r_2}} \|M_w V\|_F^2 \quad \text{subject to} \quad V^\top V = I_{r_2}, \quad V_1^\top V = 0. \quad (22)$$

Because the columns of V_2 live in $\text{range}(P_1)$, they are already orthogonal to V_1 by construction, so the full hybrid basis

$$\widehat{V}_w := [V_1, V_2] \in \mathbb{R}^{d \times r} \quad (23)$$

satisfies $\widehat{V}_w^\top \widehat{V}_w = I_r$.

Practical computation of Step 2. The top r_2 right singular vectors of $M_w P_1$ can be computed without forming $M_w P_1$ explicitly. Let

$$G := M_w^\top M_w = B^\top B + A_w^\top A_w. \quad (24)$$

Since $P_1 = I - V_1 V_1^\top$ and $V_1^\top V_1 = I_{r_1}$,

$$(M_w P_1)^\top (M_w P_1) = P_1 G P_1. \quad (25)$$

The right singular vectors of $M_w P_1$ are the eigenvectors of $P_1 G P_1$ corresponding to its largest eigenvalues. Two implementations are convenient.

Projected Lanczos. Apply Lanczos or block-Lanczos iteration to the implicit operator $x \mapsto P_1(Gx)$ starting from a random $\text{range}(P_1)$ -vector. Each matvec uses two matvecs with M_w (one with A_w and one with B , or alternatively one with B stored in the form $\Sigma V^\top$) and one rank- r_1 correction by V_1 . Extract the top r_2 Ritz pairs.

Deflated direct SVD. When $(s+k)+r$ is small enough to form M_w explicitly, compute the thin SVD of M_w , project the right singular vectors by P_1 , and orthonormalize. This is simpler to implement but costlier in the streaming regime.

Assembling the retained right basis. The hybrid retained right basis is

$$\widehat{V}_w = [V_1, V_2] \in \mathbb{R}^{d \times r}, \quad \widehat{V}_w^\top \widehat{V}_w = I_r. \quad (26)$$

The combined-score columns V_1 cover entropy-corrected leverage directions; the singular columns V_2 cover the largest remaining quadratic-energy directions of the sketch-plus-window.

Operator extraction from $\widehat{V}_w$. The remaining extraction is identical to the combined-score method. Form

$$Y_w := M_w \widehat{V}_w = [M_w V_1, M_w V_2], \quad U_{:,w} := \text{orth}(Y_w), \quad (27)$$

and compute the SVD of the left-projected restricted operator:

$$U_{:,w}^\top M_w = U_{\text{small},w} \Sigma_w V_w^\top. \quad (28)$$

The sketch carried into the next window is

$$B^{(w)} := \Sigma_w V_w^\top \in \mathbb{R}^{r \times d}. \quad (29)$$

The row memory $R^{(w)}$ is updated from raw rows seen through window w , not from sketch rows, exactly as in the combined-score method.

Step-2 efficiency in the stored-basis form. If the previous sketch is carried as

$$B = SV^\top, \quad V = [v_1, \dots, v_r], \quad S = \text{diag}(s_1, \dots, s_r), \quad (30)$$

then

$$B^\top B = VS^2V^\top, \quad G = VS^2V^\top + A_w^\top A_w. \quad (31)$$

A matvec with G costs one matvec with A_w , one matvec with $A_w^\top$, and two rank- r inner products with V . Composing with the projection P_1 adds one rank- r_1 inner product per matvec. Hence one Lanczos step on $P_1 G P_1$ has cost dominated by the two matvecs with A_w .

First-window special case. If $B = 0$, $R = \emptyset$, $s = 0$, and $n_{\text{seen}} = 0$, then $M_w = A_w$, $G = A_w^\top A_w$, and the hybrid basis is

$$V_1 := [v_1, \dots, v_{r_1}], \quad V_2 := \text{top } r_2 \text{ right singular vectors of } A_w(I - V_1 V_1^\top), \quad (32)$$

where the v_j solve the first-window combined-score problem

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j} \|A_1 v\|_2^2 \|A_1 v\|_4^{4c_1} \|A_1 v\|_2^{-4c_1}, \quad c_1 = \frac{\log(k/n)}{2 \log k}. \quad (33)$$

The retained basis is again $\widehat{V}_1 = [V_1, V_2]$, and the extraction uses $Y_1 = A_1 \widehat{V}_1$, $U_{:,1} = \text{orth}(Y_1)$, $U_{:,1}^\top A_1 = U_{\text{small},1} \Sigma_1 V_1^\top$, and $B^{(1)} = \Sigma_1 V_1^\top$.

Why orthogonalize the singular vectors against V_1 . The combined-score directions in V_1 already lie in $\mathbb{R}^d$ and are not in general singular vectors of M_w . Taking the raw top right singular vectors of M_w would typically produce columns that overlap with $\text{span}(V_1)$, wasting budget and making $\hat{V}_w$ near-singular. Projecting M_w by $P_1 = I - V_1 V_1^\top$ before taking its top singular vectors ensures the two parts of $\hat{V}_w$ are orthogonal and that V_2 captures energy not already represented by V_1 .

Projecting M_w on the right (that is, restricting to $\text{range}(P_1)$) rather than on the left preserves the interpretation: $\|M_w V_2\|_F^2$ is the largest Frobenius energy that can be collected by r_2 right directions orthogonal to V_1 . This is the right quantity to maximize when the sketch $B^{(w)} = \Sigma_w V_w^\top$ is meant to preserve $\|M_w v\|_2$ along retained directions.

Relation to pure variants. If $r_1 = r$ and $r_2 = 0$, the method reduces to the combined-score method of the previous note. If $r_1 = 0$ and $r_2 = r$, the method reduces to the standard Frequent Directions / incremental-SVD carry: the retained sketch is the top- r right singular factor of M_w , with no entropy correction. Intermediate splits interpolate between these two extremes and let the sketch simultaneously track leverage-heavy directions and leading-energy directions.

Choosing the split adaptively. A simple adaptive rule is as follows. After solving for each combined-score direction v_j , record the relative score gain

$$\Delta_j := \frac{\text{score}_w(v_j)}{\max_{i < j} \text{score}_w(v_i)}. \quad (34)$$

Stop the score phase at the first j with $\Delta_j \leq \tau$ for a prescribed tolerance $\tau \in (0, 1)$, set $r_1 := j - 1$ and $r_2 := r - r_1$. This keeps only the combined-score directions that are noticeably distinct from the earlier ones and lets the remaining budget go to singular directions, which are cheap to compute.

Practical remarks. The combined-score step remains nonconvex, so multiple restarts are useful for each v_j . The singular-direction step is convex in the eigenvector-of- $P_1 G P_1$ sense and usually converges in a handful of Lanczos iterations. Numerically tiny denominators inside the score (when $\|C_w v\|_2$, $\|C_w v\|_4$, or $E_w(v)$ is close to zero) should be guarded as in the combined-score method. When $r_2 > 0$, the sketch $B^{(w)} = \Sigma_w V_w^\top$ retains at least the top r_2 singular energy of the sketch-plus-window minus any energy already in $\text{span}(V_1)$, so the hybrid method inherits the Frequent Directions tail-error guarantee on the projected operator, while the entropy-corrected V_1 part targets the sampling problems that a pure singular-value sketch ignores.